01 Discrete Structures 200Q

Topic 1 Discrete Structures - 200 MCQs

Syllabus: Set Theory (Sets, Relations, Functions, Pigeonhole, Inclusion-Exclusion, Equivalence/Partial Orderings, Counting, Probability, Information measures), Graph (walks, paths, trails, connected/regular/bipartite, cycles/circuits, trees/rooted/spanning, eccentricity/radius/diameter/centre, Hamiltonian/Eulerian, Planar), Groups (Finite fields, Error correcting/detecting codes). Distribution: ~70 set theory/counting, ~100 graph/tree, ~30 groups/coding.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

---

Q1. In a survey, |A|=52, |B|=27, |A ? B|=22. What is |A ? B|?

A. 59

Q2. In a survey, |A|=57, |B|=47, |A ? B|=18. What is |A ? B|?

A. 86

Q3. In a survey, |A|=59, |B|=28, |A ? B|=15. What is |A ? B|?

A. 71

Q4. In a survey, |A|=27, |B|=25, |A ? B|=7. What is |A ? B|?

A. 45

Q5. In a survey, |A|=27, |B|=45, |A ? B|=15. What is |A ? B|?

A. 57

Q6. In a survey, |A|=42, |B|=28, |A ? B|=7. What is |A ? B|?

A. 63

Q7. In a survey, |A|=28, |B|=22, |A ? B|=15. What is |A ? B|?

A. 35

Q8. In a survey, |A|=29, |B|=42, |A ? B|=7. What is |A ? B|?

A. 64

Q9. In a survey, |A|=15, |B|=44, |A ? B|=8. What is |A ? B|?

A. 50

Q10. In a survey, |A|=38, |B|=60, |A ? B|=28. What is |A ? B|?

A. 69

Q11. For sets A,B,C: |A|=21, |B|=33, |C|=25, |A?B|=9, |B?C|=6, |A?C|=7, |A?B?C|=5. Find |A?B?C|.

A. 62

Q12. For sets A,B,C: |A|=25, |B|=35, |C|=36, |A?B|=9, |B?C|=7, |A?C|=5, |A?B?C|=2. Find |A?B?C|.

A. 75

Q13. For sets A,B,C: |A|=37, |B|=34, |C|=23, |A?B|=12, |B?C|=7, |A?C|=7, |A?B?C|=3. Find |A?B?C|.

A. 72

Q14. For sets A,B,C: |A|=24, |B|=29, |C|=30, |A?B|=11, |B?C|=6, |A?C|=5, |A?B?C|=6. Find |A?B?C|.

A. 70

Q15. For sets A,B,C: |A|=26, |B|=28, |C|=30, |A?B|=6, |B?C|=9, |A?C|=7, |A?B?C|=4. Find |A?B?C|.

A. 64

Q16. For sets A,B,C: |A|=23, |B|=36, |C|=30, |A?B|=8, |B?C|=6, |A?C|=9, |A?B?C|=3. Find |A?B?C|.

A. 67

Q17. For sets A,B,C: |A|=39, |B|=34, |C|=40, |A?B|=5, |B?C|=10, |A?C|=8, |A?B?C|=4. Find |A?B?C|.

A. 95

Q18. For sets A,B,C: |A|=37, |B|=38, |C|=37, |A?B|=10, |B?C|=10, |A?C|=5, |A?B?C|=3. Find |A?B?C|.

A. 90

Q19. For sets A,B,C: |A|=32, |B|=32, |C|=40, |A?B|=5, |B?C|=8, |A?C|=6, |A?B?C|=3. Find |A?B?C|.

A. 91

Q20. For sets A,B,C: |A|=24, |B|=30, |C|=23, |A?B|=10, |B?C|=5, |A?C|=12, |A?B?C|=3. Find |A?B?C|.

A. 54

Q21. If a set S has 4 elements, what is the cardinality of its power set P(S)?

A. 16

Q22. Number of binary relations from a set A with 3 elements to a set B with 4 elements is?

A. 2^12 =

Q23. Number of all functions from a set A of size 2 to a set B of size 5 is?

A. 5^2 = 2

Q24. Number of equivalence relations on an 3-element set is?

A. 8

Q25. Number of bijective functions (permutations) on an 6-element set is?

A. 6^6 = 4

Q26. If a set S has 5 elements, what is the cardinality of its power set P(S)?

A. 25

Q27. Number of binary relations from a set A with 4 elements to a set B with 2 elements is?

A. 4^2 = 1

Q28. Number of all functions from a set A of size 4 to a set B of size 4 is?

A. 16

Q29. Number of equivalence relations on an 3-element set is?

A. 6

Q30. Number of bijective functions (permutations) on an 3-element set is?

A. 8

Q31. By the pigeonhole principle, among 50 integers, at least how many have the same remainder modulo 7?

A. 8

Q32. In how many ways can 3 distinct prizes be distributed among 6 students if each student can receive any number of prizes?

A. P(6,3) =

Q33. The value of C(11,2) is?

A. 53

Q34. The value of P(9,2) (permutations without repetition) is?

A. 81

Q35. By the pigeonhole principle, among 26 integers, at least how many have the same remainder modulo 5?

A. 2

Q36. In how many ways can 4 distinct prizes be distributed among 6 students if each student can receive any number of prizes?

A. P(6,4) =

Q37. The value of C(15,3) is?

A. 2730

Q38. The value of P(11,4) (permutations without repetition) is?

A. 7924

Q39. By the pigeonhole principle, among 50 integers, at least how many have the same remainder modulo 7?

A. 9

Q40. In how many ways can 4 distinct prizes be distributed among 11 students if each student can receive any number of prizes?

A. P(11,4)

Q41. A fair coin is tossed 3 times. Probability of exactly 2 heads is?

A. 1/4

Q42. Two dice are thrown. Probability that sum is 9 is?

A. 1/9

Q43. A card is drawn from a deck. Probability it is an ace is?

A. 1/13

Q44. If P(A)=0.4, P(B)=0.5 and A,B independent, P(A?B) is?

A. 0.10

Q45. Entropy (in bits) of a binary source with P(0)=P(1)=1/2 is?

A. 2 bits

Q46. Entropy of a certain event (p=1) is?

A. Infinity

Q47. If H(X)=2 bits, it means on average how many yes/no questions are needed?

A. 0.5

Q48. A relation R is a partial order if it is?

A. Reflexi

Q49. An equivalence relation must be?

A. Antisyn

Q50. If f: A?B is both injective and surjective, it is?

A. Only su

Q51. How many onto (surjective) functions exist from a 5-element set to a 2-element set?

A. 30

Q52. In how many ways can the letters of the word 'COMPUTER' (8 distinct letters) be arranged?

A. 2^8 = 2

Q53. For a binary memoryless source with probabilities (1/8, 7/8), the information content of the less likely symbol is greater than?

A. 1 bit

Q54. How many onto (surjective) functions exist from a 4-element set to a 2-element set?

A. 14

Q55. In how many ways can the letters of the word 'COMPUTER' (8 distinct letters) be arranged?

A. 8^8

Q56. For a binary memoryless source with probabilities (1/8, 7/8), the information content of the less likely symbol is greater than?

A. Exactly

Q57. How many onto (surjective) functions exist from a 4-element set to a 2-element set?

A. 16

Q58. In how many ways can the letters of the word 'COMPUTER' (8 distinct letters) be arranged?

A. 2^8 = 2

Q59. For a binary memoryless source with probabilities (1/8, 7/8), the information content of the less likely symbol is greater than?

A. 0 bit

Q60. How many onto (surjective) functions exist from a 7-element set to a 2-element set?

A. 126

Q61. Mutual information I(X;Y) ? 0 with equality iff?

A. X and Y

Q62. For independent X,Y, H(X,Y) equals?

A. max(H(

Q63. Channel capacity is defined as?

A. Maximu

Q64. Huffman coding achieves?

A. Fixed-le

Q65. If A?B then |A?B| equals?

A. |A|

Q66. Principle of inclusion-exclusion for 2 sets is?

A. |A?B|=|

Q67. The number of subsets of size k from n elements is?

A. C(n,k)

Q68. Bayes' theorem relates?

A. Union t

Q69. For a fair 6-faced die, entropy is?

A. 1 bit

Q70. Pigeonhole principle: 13 people imply at least two born in the same?

A. Year

Q71. Number of edges in the complete graph K7 is?

A. 22

Q72. Number of edges in complete bipartite graph K2,2 is?

A. 4

Q73. A tree with 11 vertices has how many edges?

A. 55

Q74. By Cayley's formula, number of labelled spanning trees of K6 is n^(n-2). Its value for n=6 is?

A. 7776

Q75. A connected graph is Eulerian iff?

A. Every v

Q76. A graph has an Euler trail (but not circuit) iff?

A. Exactly

Q77. Maximum number of edges in a simple planar graph with 5 vertices (n?3) is bounded by 3n-6. The bound is?

A. 9

Q78. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K6 or K

Q79. A bipartite graph cannot contain?

A. Isolated

Q80. Handshaking lemma states sum of degrees equals?

A. |E|

Q81. Eccentricity of a vertex v is defined as?

A. max dis

Q82. Radius of a graph is?

A. Maximu

Q83. Diameter of the path graph P6 (n vertices) is?

A. 5

Q84. Centre of a tree consists of?

A. Always

Q85. A full m-ary tree (each internal node has 3 children) of height 2 has at most how many leaves?

A. 6

Q86. A Hamiltonian cycle visits?

A. Only od

Q87. A 4-regular graph on 8 vertices has how many edges?

A. 8

Q88. A rooted tree in which every internal node has at most 2 children is called?

A. Ternary

Q89. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Eulerian

Q90. Which statement about trees is FALSE?

A. A tree c

Q91. Number of edges in the complete graph K6 is?

A. 15

Q92. Number of edges in complete bipartite graph K4,4 is?

A. 8

Q93. A tree with 5 vertices has how many edges?

A. 6

Q94. By Cayley's formula, number of labelled spanning trees of K4 is n^(n-2). Its value for n=4 is?

A. 16

Q95. A connected graph is Eulerian iff?

A. Every v

Q96. A graph has an Euler trail (but not circuit) iff?

A. It is dis

Q97. Maximum number of edges in a simple planar graph with 7 vertices (n?3) is bounded by 3n-6. The bound is?

A. 21

Q98. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K6 or K

Q99. A bipartite graph cannot contain?

A. A span

Q100. Handshaking lemma states sum of degrees equals?

A. |V|

Q101. Eccentricity of a vertex v is defined as?

A. max dis

Q102. Radius of a graph is?

A. Maximu

Q103. Diameter of the path graph P8 (n vertices) is?

A. 8

Q104. Centre of a tree consists of?

A. Always

Q105. A full m-ary tree (each internal node has 2 children) of height 3 has at most how many leaves?

A. 5

Q106. A Hamiltonian cycle visits?

A. Every e

Q107. A 5-regular graph on 10 vertices has how many edges?

A. 26

Q108. A rooted tree in which every internal node has at most 2 children is called?

A. Binary

Q109. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Depth-f

Q110. Which statement about trees is FALSE?

A. A tree i

Q111. Number of edges in the complete graph K10 is?

A. 46

Q112. Number of edges in complete bipartite graph K4,2 is?

A. 9

Q113. A tree with 11 vertices has how many edges?

A. 55

Q114. By Cayley's formula, number of labelled spanning trees of K4 is n^(n-2). Its value for n=4 is?

A. 16

Q115. A connected graph is Eulerian iff?

A. Every v

Q116. A graph has an Euler trail (but not circuit) iff?

A. All verti

Q117. Maximum number of edges in a simple planar graph with 5 vertices (n?3) is bounded by 3n-6. The bound is?

A. 10

Q118. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K5 or K

Q119. A bipartite graph cannot contain?

A. An odd

Q120. Handshaking lemma states sum of degrees equals?

A. 2|E|

Q121. Eccentricity of a vertex v is defined as?

A. sum of

Q122. Radius of a graph is?

A. Maximu

Q123. Diameter of the path graph P7 (n vertices) is?

A. 7

Q124. Centre of a tree consists of?

A. Always

Q125. A full m-ary tree (each internal node has 4 children) of height 3 has at most how many leaves?

A. 81

Q126. A Hamiltonian cycle visits?

A. Only od

Q127. A 4-regular graph on 8 vertices has how many edges?

A. 4

Q128. A rooted tree in which every internal node has at most 2 children is called?

A. Bipartit

Q129. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Euleria

Q130. Which statement about trees is FALSE?

A. A tree i

Q131. Number of edges in the complete graph K7 is?

A. 21

Q132. Number of edges in complete bipartite graph K2,2 is?

A. 4

Q133. A tree with 6 vertices has how many edges?

A. 6

Q134. By Cayley's formula, number of labelled spanning trees of K3 is n^(n-2). Its value for n=3 is?

A. 6

Q135. A connected graph is Eulerian iff?

A. Every v

Q136. A graph has an Euler trail (but not circuit) iff?

A. It is dis

Q137. Maximum number of edges in a simple planar graph with 7 vertices (n?3) is bounded by 3n-6. The bound is?

A. 21

Q138. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K4 or K

Q139. A bipartite graph cannot contain?

A. A spann

Q140. Handshaking lemma states sum of degrees equals?

A. |V|

Q141. Eccentricity of a vertex v is defined as?

A. min dis

Q142. Radius of a graph is?

A. Maximu

Q143. Diameter of the path graph P8 (n vertices) is?

A. 8

Q144. Centre of a tree consists of?

A. All leav

Q145. A full m-ary tree (each internal node has 2 children) of height 3 has at most how many leaves?

A. 5

Q146. A Hamiltonian cycle visits?

A. Every v

Q147. A 3-regular graph on 6 vertices has how many edges?

A. 6

Q148. A rooted tree in which every internal node has at most 2 children is called?

A. Spanni

Q149. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Shortes

Q150. Which statement about trees is FALSE?

A. A tree v

Q151. Number of edges in the complete graph K5 is?

A. 11

Q152. Number of edges in complete bipartite graph K3,3 is?

A. 36

Q153. A tree with 7 vertices has how many edges?

A. 21

Q154. By Cayley's formula, number of labelled spanning trees of K4 is n^(n-2). Its value for n=4 is?

A. 64

Q155. A connected graph is Eulerian iff?

A. Every v

Q156. A graph has an Euler trail (but not circuit) iff?

A. Exactly

Q157. Maximum number of edges in a simple planar graph with 4 vertices (n?3) is bounded by 3n-6. The bound is?

A. 8

Q158. Kuratowski's theorem states a graph is planar iff it contains no subdivision of?

A. K6 or K

Q159. A bipartite graph cannot contain?

A. Isolated

Q160. Handshaking lemma states sum of degrees equals?

A. |E|

Q161. Eccentricity of a vertex v is defined as?

A. min dis

Q162. Radius of a graph is?

A. Maximu

Q163. Diameter of the path graph P9 (n vertices) is?

A. 4

Q164. Centre of a tree consists of?

A. Always

Q165. A full m-ary tree (each internal node has 4 children) of height 2 has at most how many leaves?

A. 16

Q166. A Hamiltonian cycle visits?

A. Only th

Q167. A 4-regular graph on 8 vertices has how many edges?

A. 4

Q168. A rooted tree in which every internal node has at most 2 children is called?

A. Ternary

Q169. Breadth-first search (BFS) of a graph yields which type of tree/ordering?

A. Shortes

Q170. Which statement about trees is FALSE?

A. A tree c

Q171. In any group, the identity element is?

A. Never u

Q172. Lagrange's theorem states that for finite group G and subgroup H?

A. |H| > |G

Q173. Number of elements in the finite field GF(7^3) is?

A. 343

Q174. Characteristic of a finite field GF(p^n) is?

A. 0

Q175. A linear code with minimum Hamming distance d=5 can correct up to how many errors?

A. 2

Q176. A code with minimum distance d can detect up to how many errors?

A. d-1

Q177. For a Hamming code with 3 parity bits, maximum message length satisfies 2^r ? m+r+1. For r=3, total code length n=2^r-1 equals?

A. 8

Q178. The set of all n-bit vectors under XOR forms?

A. A non-a

Q179. A generator matrix G of a linear (n,k) code has dimensions?

A. k × n

Q180. Cyclic codes are most efficiently encoded/decoded using?

A. Shift re

Q181. In any group, the identity element is?

A. Never u

Q182. Lagrange's theorem states that for finite group G and subgroup H?

A. |H| > |G

Q183. Number of elements in the finite field GF(3^3) is?

A. 6

Q184. Characteristic of a finite field GF(p^n) is?

A. p^n

Q185. A linear code with minimum Hamming distance d=3 can correct up to how many errors?

A. 1

Q186. A code with minimum distance d can detect up to how many errors?

A. d

Q187. For a Hamming code with 4 parity bits, maximum message length satisfies 2^r ? m+r+1. For r=4, total code length n=2^r-1 equals?

A. 16

Q188. The set of all n-bit vectors under XOR forms?

A. Only a

Q189. A generator matrix G of a linear (n,k) code has dimensions?

A. n × n

Q190. Cyclic codes are most efficiently encoded/decoded using?

A. Shift re

Q191. In any group, the identity element is?

A. Always

Q192. Lagrange's theorem states that for finite group G and subgroup H?

A. |H| divi

Q193. Number of elements in the finite field GF(3^2) is?

A. 9

Q194. Characteristic of a finite field GF(p^n) is?

A. n

Q195. A linear code with minimum Hamming distance d=5 can correct up to how many errors?

A. 5

Q196. A code with minimum distance d can detect up to how many errors?

A. d-1

Q197. For a Hamming code with 5 parity bits, maximum message length satisfies 2^r ? m+r+1. For r=5, total code length n=2^r-1 equals?

A. 32

Q198. The set of all n-bit vectors under XOR forms?

A. An abe

Q199. A generator matrix G of a linear (n,k) code has dimensions?

A. k × n

Q200. Cyclic codes are most efficiently encoded/decoded using?

A. K-maps

ANSWER KEY

1-B, 2-A, 3-C, 4-A, 5-A, 6-A, 7-A, 8-A, 9-D, 10-D

11-A, 12-D

02 Computer Arithmetic 200Q

Topic 2 Computer Arithmetic & Digital Logic - 200 MCQs

Syllabus: Propositional/Predicate Logic, WFF, Satisfiability/Tautology, Logic Families TTL/ECL/CMOS, Boolean algebra & minimization (K-map, Quine-McCluskey), Flip-flops, race, combinational/sequential design, Number systems (Octal/Hex/Decimal/Binary), 1's/2's complement arithmetic, Floating point (IEEE 754). Distribution: ~50 logic, ~100 digital circuits, ~50 arithmetic.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

---

Q1. Which of the following is a tautology?

A. p ? ¬p

Q2. Which formula is unsatisfiable (contradiction)?

A. p ? ¬p

Q3. A wff is satisfiable iff?

A. It is fals

Q4. De Morgan's law states ¬(p ? q) equals?

A. p ? q

Q5. p ? q is logically equivalent to?

A. ¬q ? p

Q6. Which logic family has the highest speed?

A. TTL

Q7. Which logic family has lowest static power dissipation?

A. CMOS

Q8. TTL stands for?

A. Transis

Q9. ECL's key characteristic is?

A. Saturat

Q10. CMOS inverter in steady state draws?

A. Near-ze

Q11. Predicate ?x P(x) is true when?

A. P holds

Q12. ?x ?y R(x,y) versus ?y ?x R(x,y): the former?

A. Is alway

Q13. Skolemization is used to?

A. Minimiz

Q14. A valid argument's conclusion must be true when?

A. At least

Q15. Resolution principle applies to formulas in?

A. Horn fo

Q16. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q17. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q18. Fan-out of a logic gate refers to?

A. Power

Q19. Noise margin is highest (among classic families) in?

A. CMOS

Q20. A tautology with n variables is true in how many rows of truth table?

A. 4-1 row

Q21. Which of the following is logically equivalent to ¬(p ? q)?

A. ¬p ? ¬q

Q22. The dual of Boolean expression A·(B+C) is?

A. (A+B)·C

Q23. Fan-out of a logic gate refers to?

A. Numbe

Q24. Noise margin is highest (among classic families) in?

A. ECL

Q25. A tautology with n variables is true in how many rows of truth table?

A. 16-1 ro

Q26. Which of the following is logically equivalent to ¬(p ? q)?

A. ¬p ? ¬q

Q27. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q28. Fan-out of a logic gate refers to?

A. Numbe

Q29. Noise margin is highest (among classic families) in?

A. ECL

Q30. A tautology with n variables is true in how many rows of truth table?

A. All 16 r

Q31. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q32. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q33. Fan-out of a logic gate refers to?

A. Propag

Q34. Noise margin is highest (among classic families) in?

A. I2L

Q35. A tautology with n variables is true in how many rows of truth table?

A. 2 rows

Q36. Which of the following is logically equivalent to ¬(p ? q)?

A. ¬p ? ¬q

Q37. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q38. Fan-out of a logic gate refers to?

A. Numbe

Q39. Noise margin is highest (among classic families) in?

A. CMOS

Q40. A tautology with n variables is true in how many rows of truth table?

A. All 4 ro

Q41. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q42. The dual of Boolean expression A·(B+C) is?

A. A·B+C

Q43. Fan-out of a logic gate refers to?

A. Propag

Q44. Noise margin is highest (among classic families) in?

A. ECL

Q45. A tautology with n variables is true in how many rows of truth table?

A. 2 rows

Q46. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q47. The dual of Boolean expression A·(B+C) is?

A. A+(B·C

Q48. Fan-out of a logic gate refers to?

A. Power

Q49. Noise margin is highest (among classic families) in?

A. Schottk

Q50. A tautology with n variables is true in how many rows of truth table?

A. All 16 r

Q51. An n-variable K-map (n=4) has how many cells?

A. 8

Q52. In a K-map, groups must contain cells in powers of?

A. 2 (1,2,4

Q53. Quine-McCluskey method is used for?

A. Floating

Q54. An essential prime implicant is one that?

A. Covers

Q55. A 4:1 multiplexer has how many select lines?

A. 4

Q56. A 4-to-16 decoder has how many output lines?

A. 16

Q57. Race-around condition occurs in?

A. D flip-fl

Q58. Master-slave JK flip-flop was introduced to overcome?

A. Fan-in

Q59. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = J

Q60. Characteristic equation of JK flip-flop is?

A. Q+ = T

Q61. A mod-8 ripple counter requires at least how many flip-flops?

A. 8

Q62. A combinational circuit's output depends on?

A. Presen

Q63. A sequential circuit's output depends on?

A. Tempe

Q64. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1010

Q65. The Boolean function F=A+A'B equals (absorption)?

A. A

Q66. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 1,1,1

Q67. A half-adder has inputs/outputs?

A. 2 inputs

Q68. A full-adder can be built from how many half-adders + OR?

A. 1 half-a

Q69. Propagation delay in ripple-carry adder grows as?

A. O(n) wi

Q70. Edge-triggered flip-flop samples input on?

A. Entire H

Q71. An n-variable K-map (n=3) has how many cells?

A. 8

Q72. In a K-map, groups must contain cells in powers of?

A. 3

Q73. Quine-McCluskey method is used for?

A. Encodi

Q74. An essential prime implicant is one that?

A. Covers

Q75. A 2:1 multiplexer has how many select lines?

A. 2

Q76. A 4-to-16 decoder has how many output lines?

A. 5

Q77. Race-around condition occurs in?

A. ROM

Q78. Master-slave JK flip-flop was introduced to overcome?

A. Race-a

Q79. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = D

Q80. Characteristic equation of JK flip-flop is?

A. Q+ = J

Q81. A mod-10 ripple counter requires at least how many flip-flops?

A. 5

Q82. A combinational circuit's output depends on?

A. Presen

Q83. A sequential circuit's output depends on?

A. Tempe

Q84. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1110

Q85. The Boolean function F=A+A'B equals (absorption)?

A. A'B

Q86. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2

Q87. A half-adder has inputs/outputs?

A. 2 inputs

Q88. A full-adder can be built from how many half-adders + OR?

A. 2 half-a

Q89. Propagation delay in ripple-carry adder grows as?

A. O(log n

Q90. Edge-triggered flip-flop samples input on?

A. Entire H

Q91. An n-variable K-map (n=2) has how many cells?

A. 4

Q92. In a K-map, groups must contain cells in powers of?

A. 2 (1,2,4

Q93. Quine-McCluskey method is used for?

A. Designi

Q94. An essential prime implicant is one that?

A. Covers

Q95. A 2:1 multiplexer has how many select lines?

A. 2

Q96. A 3-to-8 decoder has how many output lines?

A. 8

Q97. Race-around condition occurs in?

A. JK flip-f

Q98. Master-slave JK flip-flop was introduced to overcome?

A. Clock s

Q99. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = Q

Q100. Characteristic equation of JK flip-flop is?

A. Q+ = S

Q101. A mod-16 ripple counter requires at least how many flip-flops?

A. 5

Q102. A combinational circuit's output depends on?

A. Clock c

Q103. A sequential circuit's output depends on?

A. Power

Q104. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 0100

Q105. The Boolean function F=A+A'B equals (absorption)?

A. A'B

Q106. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 3,2,1

Q107. A half-adder has inputs/outputs?

A. 3 inputs

Q108. A full-adder can be built from how many half-adders + OR?

A. 4 half-a

Q109. Propagation delay in ripple-carry adder grows as?

A. O(log n

Q110. Edge-triggered flip-flop samples input on?

A. Entire L

Q111. An n-variable K-map (n=4) has how many cells?

A. 16

Q112. In a K-map, groups must contain cells in powers of?

A. 3

Q113. Quine-McCluskey method is used for?

A. Floating

Q114. An essential prime implicant is one that?

A. Is alwa

Q115. A 2:1 multiplexer has how many select lines?

A. 2

Q116. A 4-to-16 decoder has how many output lines?

A. 8

Q117. Race-around condition occurs in?

A. D flip-fl

Q118. Master-slave JK flip-flop was introduced to overcome?

A. Power

Q119. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = Q

Q120. Characteristic equation of JK flip-flop is?

A. Q+ = D

Q121. A mod-8 ripple counter requires at least how many flip-flops?

A. 8

Q122. A combinational circuit's output depends on?

A. Presen

Q123. A sequential circuit's output depends on?

A. Presen

Q124. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1010

Q125. The Boolean function F=A+A'B equals (absorption)?

A. A

Q126. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2

Q127. A half-adder has inputs/outputs?

A. 3 inputs

Q128. A full-adder can be built from how many half-adders + OR?

A. 3 half-a

Q129. Propagation delay in ripple-carry adder grows as?

A. O(log n

Q130. Edge-triggered flip-flop samples input on?

A. Active

Q131. An n-variable K-map (n=3) has how many cells?

A. 8

Q132. In a K-map, groups must contain cells in powers of?

A. 3

Q133. Quine-McCluskey method is used for?

A. Designi

Q134. An essential prime implicant is one that?

A. Is alway

Q135. A 2:1 multiplexer has how many select lines?

A. 1

Q136. A 3-to-8 decoder has how many output lines?

A. 8

Q137. Race-around condition occurs in?

A. JK flip-f

Q138. Master-slave JK flip-flop was introduced to overcome?

A. Fan-in

Q139. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = J

Q140. Characteristic equation of JK flip-flop is?

A. Q+ = S

Q141. A mod-8 ripple counter requires at least how many flip-flops?

A. 8

Q142. A combinational circuit's output depends on?

A. Previou

Q143. A sequential circuit's output depends on?

A. Presen

Q144. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1110

Q145. The Boolean function F=A+A'B equals (absorption)?

A. A

Q146. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2

Q147. A half-adder has inputs/outputs?

A. 2 inputs

Q148. A full-adder can be built from how many half-adders + OR?

A. 4 half-a

Q149. Propagation delay in ripple-carry adder grows as?

A. O(n) wi

Q150. Edge-triggered flip-flop samples input on?

A. Both le

Q151. Decimal 109 in 8-bit binary is?

A. 110110

Q152. Decimal 222 in hexadecimal (2 digits) is?

A. 0xDF

Q153. Decimal 86 in octal is?

A. 127

Q154. Range of signed integers in 3-bit 2's complement is?

A. -4 to +3

Q155. 1's complement of binary 01101 is?

A. 01101

Q156. In IEEE 754 single precision, exponent bias is?

A. 1023

Q157. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+7+24

Q158. In IEEE 754 double precision, exponent bias is?

A. 1023

Q159. Binary 1100 + 0111 equals decimal?

A. 18

Q160. Overflow in 2's complement addition occurs when?

A. There i

Q161. Decimal 103 in 8-bit binary is?

A. 011001

Q162. Decimal 55 in hexadecimal (2 digits) is?

A. 0x38

Q163. Decimal 87 in octal is?

A. 126

Q164. Range of signed integers in 5-bit 2's complement is?

A. -31 to 3

Q165. 1's complement of binary 10110 is?

A. 01001

Q166. In IEEE 754 single precision, exponent bias is?

A. 128

Q167. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1 sign +

Q168. In IEEE 754 double precision, exponent bias is?

A. 1023

Q169. Binary 1110 + 0010 equals decimal?

A. 17

Q170. Overflow in 2's complement addition occurs when?

A. Result i

Q171. Decimal 14 in 8-bit binary is?

A. 000111

Q172. Decimal 194 in hexadecimal (2 digits) is?

A. 0xC2

Q173. Decimal 80 in octal is?

A. 117

Q174. Range of signed integers in 7-bit 2's complement is?

A. 0 to 12

Q175. 1's complement of binary 10011 is?

A. 11111

Q176. In IEEE 754 single precision, exponent bias is?

A. 127

Q177. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1 sign -

Q178. In IEEE 754 double precision, exponent bias is?

A. 63

Q179. Binary 0010 + 0011 equals decimal?

A. 6

Q180. Overflow in 2's complement addition occurs when?

A. Operan

Q181. Decimal 79 in 8-bit binary is?

A. 010011

Q182. Decimal 73 in hexadecimal (2 digits) is?

A. 0x4A

Q183. Decimal 38 in octal is?

A. 26

Q184. Range of signed integers in 6-bit 2's complement is?

A. -32 to +

Q185. 1's complement of binary 01101 is?

A. 00000

Q186. In IEEE 754 single precision, exponent bias is?

A. 1023

Q187. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+7+24

Q188. In IEEE 754 double precision, exponent bias is?

A. 127

Q189. Binary 0011 + 0111 equals decimal?

A. 21

Q190. Overflow in 2's complement addition occurs when?

A. Result i

Q191. Decimal 145 in 8-bit binary is?

A. 100100

Q192. Decimal 25 in hexadecimal (2 digits) is?

A. 0x19

Q193. Decimal 18 in octal is?

A. 23

Q194. Range of signed integers in 7-bit 2's complement is?

A. 0 to 12

Q195. 1's complement of binary 11001 is?

A. 11111

Q196. In IEEE 754 single precision, exponent bias is?

A. 1023

Q197. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+11+2

Q198. In IEEE 754 double precision, exponent bias is?

A. 2047

Q199. Binary 0110 + 0110 equals decimal?

A. 12

Q200. Overflow in 2's complement addition occurs when?

A. Carry i

ANSWER KEY

1-A, 2-C, 3-C, 4-B, 5-C, 6-B, 7-A, 8-A, 9-D, 10-A

11-B, 12-B

03 C CPP 200Q

Topic 3 C and C++ Programming - 200 MCQs

Syllabus: C tokens/identifiers/datatypes, control structures, arrays/struct/union/string/pointers, OOP concepts (class/object/inheritance/polymorphism/overloading), C++ tokens/variables/constants/operators/control/functions/parameter passing/class/constructors/destructors/overloading/inheritance/templates/exception handling. Includes output-prediction. Distribution: ~90 C, ~30 OOP theory, ~80 C++.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

---

Q1. What is the output of: int a=5; printf("%d", a++ + ++a); assuming evaluation left-to-right with gcc giving 12? In strict ANSI C the behaviour is?

A. 12 guar

Q2. What is output of: int x=10; printf("%d %d", x++, ++x); ?

A. 10 11 g

Q3. Output of: for(i=0;i<5;i++); printf("%d",i); with i declared int i; ?

A. 0

Q4. Output of: int a[]={1,2,3,4}; printf("%d", *(a+2)); ?

A. 2

Q5. Output of: char s[]="hello"; printf("%lu", (unsigned long)strlen(s)); ?

A. 6

Q6. Output of: int x=5; if(x=0) printf("hi"); else printf("bye"); ?

A. hi bye

Q7. Output of: printf("%d", 5/2); in C (both operands int)?

A. 2

Q8. Output of: printf("%f", 5/2); ?

A. 0.0000

Q9. Output of: int n=0; while(n++<3) printf("%d",n); ?

A. 012

Q10. Output of: char c='A'; printf("%d", c+1); ?

A. B

Q11. sizeof(int) on typical 32/64-bit gcc/Linux is?

A. 1 byte

Q12. sizeof(char) in C is by definition?

A. 2 bytes

Q13. Which storage class retains value between function calls?

A. auto

Q14. Default storage class of local variables is?

A. register

Q15. Which operator has highest precedence?

A. * over +

Q16. Output of: int a=10,b=20; printf("%d", a^b); (bitwise XOR)?

A. 0

Q17. Output of: printf("%d", ~5); on 32-bit 2's complement?

A. -5

Q18. Output of: int p=5; printf("%d", p<<2); ?

A. 10

Q19. Output of: union {int i; char c[4];} u; u.i=1; printf("%d", u.c[0]); on little-endian?

A. 0

Q20. Dereferencing a NULL pointer causes?

A. Always

Q21. C tokens include?

A. Only fu

Q22. Which is NOT a valid C identifier?

A. _abc

Q23. Which is a valid constant in C?

A. Only 12

Q24. Difference between struct and union?

A. No diffe

Q25. Arrays in C are?

A. Indexed

Q26. Pointer arithmetic p+1 advances by?

A. Always

Q27. String in C is terminated by?

A. Null cha

Q28. Which function copies strings?

A. strlen

Q29. Recursion requires?

A. Global

Q30. Passing array to function in C passes?

A. By valu

Q31. Output of: int a=4,b=5; printf("%d", a*b + a/b); ?

A. 20

Q32. Output of: int i,sum=0; for(i=1;i<=7;i++) sum+=i; printf("%d",sum); ?

A. 27

Q33. Output of: printf("%d", 8 & 8); (bitwise AND)?

A. 0

Q34. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. while

Q35. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 20

Q36. Which header declares malloc/free?

A. string.h

Q37. Output of: int a=4,b=7; printf("%d", a*b + a/b); ?

A. 28

Q38. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 10

Q39. Output of: printf("%d", 1 & 10); (bitwise AND)?

A. 11

Q40. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. while

Q41. sizeof(int arr[4]) on machine with sizeof(int)=4 is?

A. 8

Q42. Which header declares malloc/free?

A. math.h

Q43. Output of: int a=6,b=6; printf("%d", a*b + a/b); ?

A. 12

Q44. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 9

Q45. Output of: printf("%d", 5 & 9); (bitwise AND)?

A. 1

Q46. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q47. sizeof(int arr[4]) on machine with sizeof(int)=4 is?

A. 4

Q48. Which header declares malloc/free?

A. string.h

Q49. Output of: int a=8,b=3; printf("%d", a*b + a/b); ?

A. 27

Q50. Output of: int i,sum=0; for(i=1;i<=6;i++) sum+=i; printf("%d",sum); ?

A. 21

Q51. Output of: printf("%d", 9 & 9); (bitwise AND)?

A. 0

Q52. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. none

Q53. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 4

Q54. Which header declares malloc/free?

A. math.h

Q55. Output of: int a=2,b=3; printf("%d", a*b + a/b); ?

A. 7

Q56. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 6

Q57. Output of: printf("%d", 6 & 1); (bitwise AND)?

A. 7

Q58. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for

Q59. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 9

Q60. Which header declares malloc/free?

A. stdlib.h

Q61. Output of: int a=7,b=3; printf("%d", a*b + a/b); ?

A. 24

Q62. Output of: int i,sum=0; for(i=1;i<=5;i++) sum+=i; printf("%d",sum); ?

A. 25

Q63. Output of: printf("%d", 10 & 4); (bitwise AND)?

A. 14

Q64. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q65. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 5

Q66. Which header declares malloc/free?

A. stdlib.h

Q67. Output of: int a=4,b=9; printf("%d", a*b + a/b); ?

A. 36

Q68. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 10

Q69. Output of: printf("%d", 6 & 8); (bitwise AND)?

A. 14

Q70. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q71. sizeof(int arr[2]) on machine with sizeof(int)=4 is?

A. 8

Q72. Which header declares malloc/free?

A. stdlib.h

Q73. Output of: int a=2,b=9; printf("%d", a*b + a/b); ?

A. 11

Q74. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 9

Q75. Output of: printf("%d", 7 & 3); (bitwise AND)?

A. 7

Q76. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for

Q77. sizeof(int arr[3]) on machine with sizeof(int)=4 is?

A. 7

Q78. Which header declares malloc/free?

A. stdlib.h

Q79. Output of: int a=3,b=6; printf("%d", a*b + a/b); ?

A. 18

Q80. Output of: int i,sum=0; for(i=1;i<=5;i++) sum+=i; printf("%d",sum); ?

A. 14

Q81. Output of: printf("%d", 7 & 1); (bitwise AND)?

A. 1

Q82. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for

Q83. sizeof(int arr[2]) on machine with sizeof(int)=4 is?

A. 6

Q84. Which header declares malloc/free?

A. math.h

Q85. Output of: int a=9,b=2; printf("%d", a*b + a/b); ?

A. 23

Q86. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 9

Q87. Output of: printf("%d", 5 & 5); (bitwise AND)?

A. 5

Q88. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q89. sizeof(int arr[3]) on machine with sizeof(int)=4 is?

A. 3

Q90. Which header declares malloc/free?

A. math.h

Q91. Encapsulation primarily achieves?

A. Data hi

Q92. Inheritance represents?

A. IS-A rel

Q93. Polymorphism in OOP means?

A. Only da

Q94. Abstraction focuses on?

A. Hiding i

Q95. A class vs object?

A. Class is

Q96. Constructor is called?

A. At obje

Q97. Destructor in C++ is denoted by?

A. Hash (#

Q98. Virtual function enables?

A. Static b

Q99. Pure virtual function makes class?

A. Concre

Q100. Function overloading requires?

A. Differer

Q101. Operator overloading cannot overload?

A. ::, ., .*,

Q102. Friend function in C++?

A. Cannot

Q103. Multiple inheritance ambiguity is resolved via?

A. Macros

Q104. HAS-A relationship is implemented via?

A. Inherita

Q105. Overriding vs overloading?

A. Overloa

Q106. Which OOP pillar binds code and data together?

A. Abstrac

Q107. Which OOP pillar enables code reuse via hierarchy?

A. Abstrac

Q108. Dynamic binding is associated with?

A. Macros

Q109. An abstract class in C++ has?

A. Only st

Q110. Which OOP pillar binds code and data together?

A. Inherita

Q111. Which OOP pillar enables code reuse via hierarchy?

A. Abstrac

Q112. Dynamic binding is associated with?

A. Virtual

Q113. An abstract class in C++ has?

A. At least

Q114. Which OOP pillar binds code and data together?

A. Abstrac

Q115. Which OOP pillar enables code reuse via hierarchy?

A. Encaps

Q116. Dynamic binding is associated with?

A. Inline fu

Q117. An abstract class in C++ has?

A. No mer

Q118. Which OOP pillar binds code and data together?

A. Inherita

Q119. Which OOP pillar enables code reuse via hierarchy?

A. Inherita

Q120. Dynamic binding is associated with?

A. Virtual

Q121. Which is correct C++ input/output?

A. cin >> x

Q122. Scope resolution operator is?

A. ->

Q123. Which header for C++ strings?

A. <string.

Q124. new/delete in C++ differ from malloc/free because?

A. They ar

Q125. Default access in C++ class is?

A. public

Q126. Default access in C++ struct is?

A. none

Q127. Copy constructor is called when?

A. Every a

Q128. Which constructor has no parameters?

A. Default

Q129. Function template allows?

A. No reus

Q130. Class template parameter is declared with?

A. overloa

Q131. Exception handling keywords are?

A. do, whi

Q132. An uncaught exception causes?

A. termina

Q133. STL vector provides?

A. Associa

Q134. Which STL container is LIFO?

A. set

Q135. Inline function suggests?

A. No retu

Q136. 'this' pointer refers to?

A. Current

Q137. Pass by reference in C++ uses?

A. * only

Q138. Function overloading resolution is?

A. Link-tim

Q139. Virtual destructor is needed when?

A. Only fo

Q140. Namespace in C++ is used to?

A. Avoid n

Q141. Output of C++: int x=6; cout << (x>10 ? "big" : "small"); ?

A. error

Q142. Output of C++: for(int i=0;i<5;i++) cout<<i; ?

A. 43210

Q143. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try alwa

Q144. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referen

Q145. Which cast is safest for polymorphism in C++?

A. const_c

Q146. Template specialization allows?

A. Custom

Q147. Output of C++: int x=19; cout << (x>10 ? "big" : "small"); ?

A. error

Q148. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 0120

Q149. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch a

Q150. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. NULL

Q151. Which cast is safest for polymorphism in C++?

A. dynami

Q152. Template specialization allows?

A. Custom

Q153. Output of C++: int x=7; cout << (x>10 ? "big" : "small"); ?

A. 0

Q154. Output of C++: for(int i=0;i<2;i++) cout<<i; ?

A. 2

Q155. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q156. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referer

Q157. Which cast is safest for polymorphism in C++?

A. const_c

Q158. Template specialization allows?

A. Faster

Q159. Output of C++: int x=13; cout << (x>10 ? "big" : "small"); ?

A. big

Q160. Output of C++: for(int i=0;i<6;i++) cout<<i; ?

A. 012345

Q161. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch a

Q162. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0

Q163. Which cast is safest for polymorphism in C++?

A. C-style

Q164. Template specialization allows?

A. No gen

Q165. Output of C++: int x=9; cout << (x>10 ? "big" : "small"); ?

A. error

Q166. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 3

Q167. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q168. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. NULL

Q169. Which cast is safest for polymorphism in C++?

A. dynami

Q170. Template specialization allows?

A. Only m

Q171. Output of C++: int x=6; cout << (x>10 ? "big" : "small"); ?

A. 0

Q172. Output of C++: for(int i=0;i<2;i++) cout<<i; ?

A. 01

Q173. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q174. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referen

Q175. Which cast is safest for polymorphism in C++?

A. const_c

Q176. Template specialization allows?

A. No gen

Q177. Output of C++: int x=11; cout << (x>10 ? "big" : "small"); ?

A. small

Q178. Output of C++: for(int i=0;i<4;i++) cout<<i; ?

A. 4

Q179. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try alwa

Q180. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 5

Q181. Which cast is safest for polymorphism in C++?

A. dynami

Q182. Template specialization allows?

A. No gen

Q183. Output of C++: int x=9; cout << (x>10 ? "big" : "small"); ?

A. big

Q184. Output of C++: for(int i=0;i<6;i++) cout<<i; ?

A. 543210

Q185. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q186. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0

Q187. Which cast is safest for polymorphism in C++?

A. reinterp

Q188. Template specialization allows?

A. Faster

Q189. Output of C++: int x=2; cout << (x>10 ? "big" : "small"); ?

A. error

Q190. Output of C++: for(int i=0;i<5;i++) cout<<i; ?

A. 43210

Q191. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch a

Q192. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referer

Q193. Which cast is safest for polymorphism in C++?

A. const_c

Q194. Template specialization allows?

A. Only m

Q195. Output of C++: int x=15; cout << (x>10 ? "big" : "small"); ?

A. big

Q196. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 0120

Q197. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try alwa

Q198. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0

Q199. Which cast is safest for polymorphism in C++?

A. const_c

Q200. Template specialization allows?

A. No gen

ANSWER KEY

1-D, 2-D, 3-C, 4-B, 5-B, 6-D, 7-A, 8-D, 9-B, 10-C

11-C, 12-C

04 TOC 200Q

Topic 4 Theory of Computation - 200 MCQs

Syllabus: Finite Automata, PDA, NFA/DPDA, non-determinism, languages accepted, Grammars, Non-computability, examples of non-computable problems. Includes DFA/NFA conversions, pumping lemma, Chomsky hierarchy, Turing machines, decidability/halting. ~200 Qs.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

---

Q1. A DFA is formally a 5-tuple (Q, ?, ?, q0, F) where ? is?

A. Q × ? ?

Q2. An NFA transition function maps to?

A. Single s

Q3. Every NFA can be converted to an equivalent DFA with at most?

A. n! state

Q4. ?-closure of a state includes?

A. Only fir

Q5. Language accepted by DFA is called?

A. Contex

Q6. Pumping lemma for regular languages states: for long enough string xyz with |y|>0, |xy|?p, then?

A. p must

Q7. Which language is NOT regular?

A. (01)*

Q8. Myhill-Nerode theorem characterizes regularity via?

A. Finite i

Q9. Minimal DFA is unique up to isomorphism and found by?

A. Remov

Q10. Regular expressions, NFA, DFA are?

A. Only R

Q11. Context-free grammars are recognized by?

A. Finite a

Q12. A DPDA is strictly weaker than?

A. NFA

Q13. Example of deterministic CFL but ambiguous grammar?

A. a^n b^n

Q14. Chomsky Normal Form productions are of type?

A. A?? on

Q15. Greibach Normal Form productions start with?

A. Termin

Q16. Chomsky hierarchy from most restricted: Regular ? CFL ? CSL ? ?

A. Finite o

Q17. Type-0 grammars correspond to?

A. PDA

Q18. Type-1 (context-sensitive) correspond to?

A. Linear

Q19. Pumping lemma for CFL uses?

A. No bou

Q20. Which language is CFL but not regular?

A. {ww | w

Q21. Ambiguous grammar means?

A. Infinite

Q22. Inherent ambiguity example?

A. (a+b)*

Q23. CYK algorithm decides membership for CFG in CNF in?

A. O(n)

Q24. Pushdown automaton uses?

A. Stack (

Q25. Turing machine adds over PDA?

A. Only st

Q26. Church-Turing thesis states?

A. P=NP

Q27. Halting problem is?

A. Regula

Q28. An example non-computable / undecidable problem?

A. Sorting

Q29. Rice's theorem says?

A. TMs alw

Q30. Decidable problems for DFA include?

A. Halting

Q31. Which CFL problem is undecidable?

A. Membe

Q32. Post Correspondence Problem (PCP) is?

A. Finite

Q33. Linear bounded automata decide?

A. Regula

Q34. A universal Turing machine?

A. Cannot

Q35. Recursively enumerable vs recursive?

A. Recurs

Q36. Complement of halting language is?

A. CFL

Q37. If L and its complement are both RE then L is?

A. Recurs

Q38. Kleene closure L* is regular if L is regular: proof uses?

A. ?-NFA

Q39. Arden's theorem solves?

A. Equatio

Q40. Mealy vs Moore machines differ in?

A. Output

Q41. The number of states in minimal DFA for language (0+1)*0(0+1){2} (k fixed suffix length) is at least?

A. 8

Q42. Consider the standard result: An NFA transition function maps to?

A. Stack s

Q43. L = {a^n b^n c^n} is?

A. Regula

Q44. If NFA has 3 states, worst-case DFA via subset construction has?

A. 7 state

Q45. Consider the standard result: Language accepted by DFA is called?

A. Recurs

Q46. Intersection of CFL with regular is?

A. Regula

Q47. Pumping length p for DFA with 3 states can be taken as?

A. 4

Q48. Consider the standard result: Myhill-Nerode theorem characterizes regularity via?

A. Stack d

Q49. L = {a^p | p prime} is?

A. Star-fre

Q50. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Consta

Q51. Consider the standard result: Context-free grammars are recognized by?

A. Turing

Q52. Empty-string membership for RE language is?

A. Always

Q53. For alphabet size 3, number of strings of length n is?

A. 3^n

Q54. Consider the standard result: Chomsky Normal Form productions are of type?

A. A?a*

Q55. Complement of CFL is?

A. Not nec

Q56. The number of states in minimal DFA for language (0+1)*0(0+1){2} (k fixed suffix length) is at least?

A. 8

Q57. Consider the standard result: Type-0 grammars correspond to?

A. Turing

Q58. L = {ww^R | w?{a,b}*} is?

A. Determ

Q59. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5^2 sta

Q60. Consider the standard result: Which language is CFL but not regular?

A. {a^n b^

Q61. Equivalence of two DFAs is decidable by?

A. Minimiz

Q62. Pumping length p for DFA with 3 states can be taken as?

A. 3

Q63. Consider the standard result: CYK algorithm decides membership for CFG in CNF in?

A. O(2^n)

Q64. Every finite language is?

A. Not reg

Q65. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Expone

Q66. Consider the standard result: Church-Turing thesis states?

A. P=NP

Q67. L = {a^n b^n c^n} is?

A. CFL

Q68. For alphabet size 3, number of strings of length n is?

A. 3+n

Q69. Consider the standard result: Rice's theorem says?

A. P=NP

Q70. Intersection of CFL with regular is?

A. Undeci

Q71. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 17

Q72. Consider the standard result: Post Correspondence Problem (PCP) is?

A. Decida

Q73. L = {a^p | p prime} is?

A. Star-fre

Q74. If NFA has 4 states, worst-case DFA via subset construction has?

A. 2^4 = 1

Q75. Consider the standard result: Recursively enumerable vs recursive?

A. Recurs

Q76. Empty-string membership for RE language is?

A. Semi-d

Q77. Pumping length p for DFA with 4 states can be taken as?

A. 3

Q78. Consider the standard result: Kleene closure L* is regular if L is regular: proof uses?

A. PDA

Q79. Complement of CFL is?

A. Always

Q80. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Consta

Q81. Consider the standard result: A DFA is formally a 5-tuple (Q, ?, ?, q0, F) where ? is?

A. Q ? Q

Q82. L = {ww^R | w?{a,b}*} is?

A. Regula

Q83. For alphabet size 5, number of strings of length n is?

A. 5*n

Q84. Consider the standard result: ?-closure of a state includes?

A. Only st

Q85. Equivalence of two DFAs is decidable by?

A. Halting

Q86. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 17

Q87. Consider the standard result: Which language is NOT regular?

A. {a^n | n

Q88. Every finite language is?

A. Undeci

Q89. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 sta

Q90. Consider the standard result: Regular expressions, NFA, DFA are?

A. Equival

Q91. L = {a^n b^n c^n} is?

A. Contex

Q92. Pumping length p for DFA with 5 states can be taken as?

A. 5

Q93. Consider the standard result: Example of deterministic CFL but ambiguous grammar?

A. Halting

Q94. Intersection of CFL with regular is?

A. Undeci

Q95. Number of productions to convert CFG with 4 productions to CNF is bounded by?

A. Expone

Q96. Consider the standard result: Chomsky hierarchy from most restricted: Regular ? CFL ? CSL ? ?

A. Finite o

Q97. L = {a^p | p prime} is?

A. Star-fre

Q98. For alphabet size 5, number of strings of length n is?

A. 5+n

Q99. Consider the standard result: Pumping lemma for CFL uses?

A. xyz onl

Q100. Empty-string membership for RE language is?

A. Regula

Q101. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 32

Q102. Consider the standard result: Inherent ambiguity example?

A. (a+b)*

Q103. Complement of CFL is?

A. Not nec

Q104. If NFA has 4 states, worst-case DFA via subset construction has?

A. 4^2 sta

Q105. Consider the standard result: Turing machine adds over PDA?

A. Only st

Q106. L = {ww^R | w?{a,b}*} is?

A. Not CF

Q107. Pumping length p for DFA with 3 states can be taken as?

A. 3

Q108. Consider the standard result: An example non-computable / undecidable problem?

A. Determ

Q109. Equivalence of two DFAs is decidable by?

A. Halting

Q110. Number of productions to convert CFG with 3 productions to CNF is bounded by?

A. Polynor

Q111. Consider the standard result: Which CFL problem is undecidable?

A. Emptin

Q112. Every finite language is?

A. Not reg

Q113. For alphabet size 5, number of strings of length n is?

A. 5*n

Q114. Consider the standard result: A universal Turing machine?

A. Simulat

Q115. L = {a^n b^n c^n} is?

A. CFL

Q116. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 7

Q117. Consider the standard result: If L and its complement are both RE then L is?

A. Contex

Q118. Intersection of CFL with regular is?

A. Not CF

Q119. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5 states

Q120. Consider the standard result: Mealy vs Moore machines differ in?

A. Input al

Q121. L = {a^p | p prime} is?

A. Regula

Q122. Pumping length p for DFA with 4 states can be taken as?

A. 5

Q123. Consider the standard result: Every NFA can be converted to an equivalent DFA with at most?

A. 2^n sta

Q124. Empty-string membership for RE language is?

A. Always

Q125. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Polynor

Q126. Consider the standard result: Pumping lemma for regular languages states: for long enough string xyz with |y|>0, |xy|?p, then?

A. xy^iz ?

Q127. Complement of CFL is?

A. Always

Q128. For alphabet size 2, number of strings of length n is?

A. 2^n

Q129. Consider the standard result: Minimal DFA is unique up to isomorphism and found by?

A. Partitio

Q130. L = {ww^R | w?{a,b}*} is?

A. Not CF

Q131. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 65

Q132. Consider the standard result: A DPDA is strictly weaker than?

A. DFA

Q133. Equivalence of two DFAs is decidable by?

A. Halting

Q134. If NFA has 4 states, worst-case DFA via subset construction has?

A. 4^2 sta

Q135. Consider the standard result: Greibach Normal Form productions start with?

A. Variabl

Q136. Every finite language is?

A. CFL on

Q137. Pumping length p for DFA with 2 states can be taken as?

A. 3

Q138. Consider the standard result: Type-1 (context-sensitive) correspond to?

A. Linear

Q139. L = {a^n b^n c^n} is?

A. Finite

Q140. Number of productions to convert CFG with 4 productions to CNF is bounded by?

A. Undeci

Q141. Consider the standard result: Ambiguous grammar means?

A. No term

Q142. Intersection of CFL with regular is?

A. Undeci

Q143. For alphabet size 3, number of strings of length n is?

A. n^3

Q144. Consider the standard result: Pushdown automaton uses?

A. Tape o

Q145. L = {a^p | p prime} is?

A. Finite

Q146. The number of states in minimal DFA for language (0+1)*0(0+1){4} (k fixed suffix length) is at least?

A. 6

Q147. Consider the standard result: Halting problem is?

A. Decidal

Q148. Empty-string membership for RE language is?

A. Regula

Q149. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5 states

Q150. Consider the standard result: Decidable problems for DFA include?

A. Ambigu

Q151. Complement of CFL is?

A. Always

Q152. Pumping length p for DFA with 4 states can be taken as?

A. 3

Q153. Consider the standard result: Linear bounded automata decide?

A. Regula

Q154. L = {ww^R | w?{a,b}*} is?

A. Regula

Q155. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Expone

Q156. Consider the standard result: Complement of halting language is?

A. Regula

Q157. Equivalence of two DFAs is decidable by?

A. Minimiz

Q158. For alphabet size 2, number of strings of length n is?

A. 2*n

Q159. Consider the standard result: Arden's theorem solves?

A. CFGs

Q160. Every finite language is?

A. CFL on

Q161. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 7

Q162. Consider the standard result: An NFA transition function maps to?

A. Single

Q163. L = {a^n b^n c^n} is?

A. Contex

Q164. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 = 4

Q165. Consider the standard result: Language accepted by DFA is called?

A. Regula

Q166. Intersection of CFL with regular is?

A. Regula

Q167. Pumping length p for DFA with 4 states can be taken as?

A. 3

Q168. Consider the standard result: Myhill-Nerode theorem characterizes regularity via?

A. Infinite

Q169. L = {a^p | p prime} is?

A. Not reg

Q170. Number of productions to convert CFG with 3 productions to CNF is bounded by?

A. Undeci

Q171. Consider the standard result: Context-free grammars are recognized by?

A. Pushdo

Q172. Empty-string membership for RE language is?

A. Semi-d

Q173. For alphabet size 5, number of strings of length n is?

A. 5+n

Q174. Consider the standard result: Chomsky Normal Form productions are of type?

A. A?BC o

Q175. Complement of CFL is?

A. Always

Q176. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 8

Q177. Consider the standard result: Type-0 grammars correspond to?

A. DFA

Q178. L = {ww^R | w?{a,b}*} is?

A. Not CF

Q179. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 sta

Q180. Consider the standard result: Which language is CFL but not regular?

A. {a^n b^

Q181. Equivalence of two DFAs is decidable by?

A. Pumpir

Q182. Pumping length p for DFA with 3 states can be taken as?

A. 4

Q183. Consider the standard result: CYK algorithm decides membership for CFG in CNF in?

A. O(n^3)

Q184. Every finite language is?

A. Undeci

Q185. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Expone

Q186. Consider the standard result: Church-Turing thesis states?

A. NFA=D

Q187. L = {a^n b^n c^n} is?

A. Regula

Q188. For alphabet size 2, number of strings of length n is?

A. 2^n

Q189. Consider the standard result: Rice's theorem says?

A. P=NP

Q190. Intersection of CFL with regular is?

A. Undeci

Q191. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 5

Q192. Consider the standard result: Post Correspondence Problem (PCP) is?

A. Finite

Q193. L = {a^p | p prime} is?

A. Star-fre

Q194. If NFA has 3 states, worst-case DFA via subset construction has?

A. 3^2 sta

Q195. Consider the standard result: Recursively enumerable vs recursive?

A. Recurs

Q196. Empty-string membership for RE language is?

A. Semi-d

Q197. Pumping length p for DFA with 4 states can be taken as?

A. 2k

Q198. Consider the standard result: Kleene closure L* is regular if L is regular: proof uses?

A. Pumpir

Q199. Complement of CFL is?

A. Always

Q200. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Expone

ANSWER KEY

1-D, 2-D, 3-C, 4-D, 5-D, 6-B, 7-C, 8-A, 9-D, 10-C

11-C, 12-B

05 Computer Organization 200Q

05 - Computer Organization - 200 MCQs (Assistant Professor Screening)

Instructions

- Q1-Q200

Q1. Whic

ANSWE

06 RDBMS SQL 200Q

06 - RDBMS & SQL - 200 MCQs (Assistant Professor Screening)

Instructions

- Q1-Q200

Q1. Entity

ANSWE

07 Data File Structures 200Q

07 - Data & File Structures - 200 MCQs (Assistant Professor Screening)

Instructions

- Q1-Q200

Q1. Array

ANSWE

08 Computer Networks 200Q

08 - Computer Networks - 200 MCQs (Assistant Professor Screening)

Instructions

- Q1-Q200

Q1. LAN s

ANSWE

09 System Software Compilers 200Q

09 - System Software & Compilers - 200 MCQs (Assistant Professor Computer Science)

Instructions

- Q1-Q200

Q1. Whic

ANSWE

10 Operating Systems Unix 200Q

10 - Operating Systems & Unix - 200 MCQs (Assistant Professor Computer Science)

Instructions

- Q1-Q200

Q1. Prima

ANSWE

11 Software Engineering 200Q

11 - Software Engineering - 200 MCQs (Assistant Professor Computer Science)

Instructions

- Q1-Q200

Q1. SDLC

ANSWER

12a Parallel Computing 50Q

AICTE Assistant Professor Screening Test - Computer Science

Topic 12

Syllabus: Parallel Virtual Machine (PVM) and Message Passing Interface (MPI) libraries and calls, Advanced architectures, Today's fastest computers.

Q1. PVM stands for:

A) Parall

Q2. MPI stands for:

A) Messa

Q3. Which of the following is a distributed-memory parallel programming standard?

A) OpenM

Q4. In Flynn's taxonomy, SISD refers to:

A) Single

Q5. GPUs are best classified as:

A) SISD

Q6. Which MPI routine must be called first in every MPI program?

A) MPI_C

Q7. Which MPI routine must be called last to cleanly exit?

A) MPI_In

Q8. In MPI, MPI_Comm_rank returns:

A) Total n

Q9. In MPI, MPI_Comm_size returns:

A) Rank o

Q10. Default communicator that includes all processes is:

A) MPI_C

Q11. Which MPI call sends a message (blocking)?

A) MPI_R

Q12. Which MPI call receives a message (blocking)?

A) MPI_S

Q13. Non-blocking send in MPI is:

A) MPI_S

Q14. MPI_Bcast is used for:

A) Point-t

Q15. MPI_Reduce is used to:

A) Reduce

Q16. MPI_Barrier does:

A) Encryp

Q17. MPI_Scatter:

A) Gather

Q18. MPI_Gather:

A) Collect

Q19. PVM was developed primarily at:

A) Micros

Q20. Which is TRUE for PVM vs MPI?

A) PVM s

Q21. In PVM, the daemon that must run on each host is called:

A) pvmd

Q22. PVM function pvm_spawn is used to:

A) Termin

Q23. PVM function pvm_send / pvm_recv are for:

A) File I/O

Q24. Amdahl's Law gives:

A) Maximu

Q25. If 90% of a program is parallelizable, with infinite processors, maximum speedup per Amdahl is:

A) 5x

Q26. Gustafson's Law (scaled speedup) argues:

A) Speedu

Q27. Speedup S(p) is defined as:

A) T_para

Q28. Efficiency E(p) is defined as:

A) S(p) / p

Q29. Which is an example of shared-memory API?

A) MPI

Q30. UMA stands for:

A) Uniform

Q31. NUMA stands for:

A) Non-Un

Q32. In a cluster, nodes are connected via:

A) High-sp

Q33. Beowulf clusters are built using:

A) Off-the-

Q34. Flynn's MIMD includes:

A) Multico

Q35. SIMD machines include:

A) Vector

Q36. Which topology has diameter log2(N) for N nodes?

A) Linear

Q37. Deadlock in message passing can occur when:

A) Two pr

Q38. MPI_TAG is used to:

A) Identify

Q39. MPI_Status object after MPI_Recv contains:

A) Source

Q40. Collective communications in MPI require:

A) All proc

Q41. Which MPI operation finds maximum value across processes?

A) MPI_R

Q42. Granularity in parallel computing refers to:

A) Ratio o

Q43. Data parallelism means:

A) Same

Q44. Task parallelism means:

A) Differen

Q45. Advanced architecture: Multi-core is:

A) Multiple

Q46. GPGPU computing commonly uses:

A) CUDA,

Q47. Today's fastest supercomputers (exascale era, e.g., Frontier, Aurora) achieve:

A) ExaFL

Q48. TOP500 list ranks supercomputers based on:

A) LINPA

Q49. Which interconnect is most common in TOP supercomputers for low latency?

A) InfiniBa

Q50. PVM supports:

A) Hetero

ANSWER KEY (12a Parallel Computing)

1-B, 2-A, 3-C, 4-A, 5-B, 6-B, 7-B, 8-B, 9-C, 10-C,

11-B, 12-E

12b Mobile Computing 200Q

12b Mobile Computing - 200 MCQs (Assistant Professor Level)

Q1. Mobile Computing framework primarily enables?

A) Fixed d

ANSWER KEY

1-B, 2-A,
23-A, 24-A
44-A, 45-A
65-A, 66-A
86-A, 87-A
105-A, 10
122-A, 12

12c Data Warehousing Mining 200Q

12c Data Warehousing & Mining - 200 MCQs (Assistant Professor Level)

Q1. Data warehouse is?

A) Subject

ANSWER KEY

1-A, 2-A,
23-A, 24-A
44-A, 45-A
65-A, 66-A
86-A, 87-A
105-A, 10
122-A, 12

12d AI 200Q

12d Artificial Intelligence - 200 MCQs (Assistant Professor Level)

Q1. AI aims to?

A) Build in

ANSWER KEY

1-A, 2-A,
23-A, 24-A
44-A, 45-A
65-A, 66-A
86-A, 87-A
105-A, 10
122-A, 12

12e E Technologies 200Q

12e E-Technologies ? 200 MCQs (Assistant Professor Level)

> Coverage: E-commerce framework / media convergence / consumer & organisation apps, e-payment (digital token / smart / credit cards, risks, design), EDI concepts / applications / legal-security-privacy / standardization / software / envelope / internet-EDI, digital libraries / data warehousing, software agents (characteristics / properties / applets / browsers), broadband telecom (frame relay / cell relay / SMDS / ATM), GIS, E-cash / E-business / ERP.

Instructions: Q1?Q200, each with options A/B/C/D. Answer Key at end.

Q1. In E-commerce framework, which layer provides multimedia content creation and distribution infrastructure?

A) Networ
business s

ANSWER KEY (12e)

Q1: B, Q2: B, Q3: D, Q4: B, Q5: A, Q6: B, Q7: A, Q8: A, Q9: A, Q10: A,

Q11: A, Q

Note: Correct option is A for all Q (distractors B/C/D). Shuffle options during practice to avoid pattern learning.

12f Windows Programming 200Q

12f Windows Programming ? 200 MCQs (Assistant Professor Level)

> Coverage: Win32 / MFC, documents & views, resources, message handling, scrolling / splitting views / docking toolbars / status bars / common dialogs, MDI, multithreading, OLE, ActiveX, ATL, network programming. Includes WinMain, message loop, MFC hierarchy (CObject ? CCmdTarget ? CWnd ? CView / CFrameWnd, CDocument).

Instructions: Q1?Q200, options A/B/C/D. Answer Key at end.

Q1. WinMain signature in Win32 is: [B]

A) int mai

ANSWER KEY (12f)

Q1: B, Q2: C, Q3: D, Q4: A, Q5: B, Q6: C, Q7: A, Q8: D, Q9: B, Q10: C,

Q11: A, Q